Junying Wang

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Education

University of Southern California, Los Angeles, CA **08/2020 – Present**
Ph.D. Candidate in Computer Science
Advisors: Prof. Ulrich Neumann and Prof. Yue Wang

University of Southern California, Los Angeles, CA **01/2018 – 05/2020**
Master of Engineering in Computer Science
Advisor: Prof. Hao Li

Ocean University of China, Qingdao, China **06/2013 – 07/2017**
Bachelor of Engineering in Computer Science

Research Interests & Skills

Research Focus	3D Vision, Computer Graphics, and Embodied AI
Research Topics	Digital Humans, Humanoid Robotics, 3D Reconstruction, Relighting, Neural Rendering, Generative AI
Programming	Python, C/C++, Java
Software & Tools	PyTorch, Claude Code/Codex, MuJoCo, Isaac Sim, OpenGL, Blender, Maya, Unity

Work Experience

Research Scientist Intern, Meta **06/2026 – Present**
Mentor: Dr. Yuting Ye

- Developing **RobotDaily**, a scalable, photorealistic real-to-sim data generator for everyday robotic tasks.
- Creating high-quality, simulation-ready 3D assets via 3DGS capture, reconstruction, inpainting, and deshading & relighting.
- Training RL policies in reconstructed scenes to generate diverse trajectories for VLA training and real-to-sim-to-real evaluation.

Research Scientist Intern, Meta **05/2025 – 12/2025**
Mentor: Dr. Yuanlu Xu

- Worked on **generalizable human 3D Gaussian splatting editing and enhancement**.
- Explored multi-layer Gaussian avatar editing and enhancement using 3D generative models.
- Contributed to one NeurIPS 2026 submission (under review).

Research Scientist Intern, Meta **06/2024 – 12/2024**
Mentor: Dr. Tony Tung

- Worked on **Dynamic Gaussian Hair (DGH)**.
- Developed a data-driven dynamic hair solver for modeling and rendering dynamic hair using 3D Gaussian representations.
- Contributed to one publication at NeurIPS 2025.

Research Scientist Intern, Adobe **09/2023 – 01/2024**
Mentor: Dr. Jae Shin Yoon

- Worked on **repurposing diffusion models for generalizable and consistent monocular human relighting**.
- Proposed a novel method that controls lighting in images and videos of humans with arbitrary body-part compositions.
- Contributed to one publication at CVPR 2025.

Research Scientist Intern, Adobe **05/2022 – 12/2022**
Mentor: Dr. Jae Shin Yoon

- Worked on **complete 3D human reconstruction from a single incomplete image**.
- Built a novel model to reconstruct complete human geometry and texture from a partial body image.
- Contributed to one publication at CVPR 2023.

Publications

- **Humanoid Everywhere: Large-Scale Real-to-Sim Environment and Data Generator for Humanoid Loco-Manipulation** **2026**
  Junying Wang, Xinjie Chen, Yunkai Zhan, Junsoo Kim, Songlin Wei, Xulang Guan, Vitor Guizilini, Chuhang Zou†, Jiageng Mao†, Yue Wang†
  Robotics: Science and Systems, RoboData Workshop (RSSW)
- **DGH: Dynamic Gaussian Hair** **2025**
  Junying Wang, Yuanlu Xu, Edith Tretschk, Ziyan Wang, Anastasia Ianina, Aljaž Božič, Ulrich Neumann, Tony Tung
  Conference on Neural Information Processing Systems (NeurIPS)
- **Comprehensive Relighting: Generalizable and Consistent Monocular Human Relighting and Harmonization** **2025**
  Junying Wang, Jingyuan Liu, Xin Sun, Krishna Kumar Singh, Zhixin Shu, He Zhang, Jimei Yang, Nanxuan Zhao, Tuanfeng Yang Wang, Simon Su Chen, Ulrich Neumann, Jae Shin Yoon
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- **Boosting Generalizability towards Zero-Shot Cross-Dataset Single-Image Indoor Depth by Meta-Initialization** **2024**
  Cho-Ying Wu, Yiqi Zhong, **Junying Wang**, Ulrich Neumann
  IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- **Complete 3D Human Reconstruction from a Single Incomplete Image** **2023**
  Junying Wang, Jae Shin Yoon, Tuanfeng Wang, Krishna Kumar Singh, Ulrich Neumann
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- **Meta-Optimization for Higher Model Generalizability in Single-Image Depth Prediction** **2023**
  Cho-Ying Wu, Yiqi Zhong, **Junying Wang**, Ulrich Neumann
  IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)

Patents

- **Systems and Methods for Generating a Relighted Image** **filed 01/2026**
  Junying Wang, Jae Shin Yoon, Jingyuan Liu, Xin Sun, Krishna Kumar Singh, Zhixin Shu, He Zhang, Jimei Yang, Yangtuanfeng Wang, Nanxuan Zhao, Simon Chen
  U.S. Patent Application No. 18/775,063
- **Complete 3D Object Reconstruction from an Incomplete Image** **filed 03/2025**
  Jae Shin Yoon, Tuanfeng Wang, Krishna Kumar Singh, **Junying Wang**, Jingwan Lu
  U.S. Patent Application No. 18/242,380

Selected Projects

Free-View Volumetric Human Body Capture from Sparse Views **05/2021 – 11/2021**
The first non-parametric human body capture system using volumetric rendering with only 2D image supervision.

- Built a novel end-to-end trainable sparse-view volumetric human body capture system for high-quality, free-view rendering.
- Proposed a novel contrastive learning approach to ensure local smoothness and continuity of the human body surface.

Contrastive Learning for Image Disentanglement **01/2020 – 06/2020**
A general model for image disentanglement trained via contrastive learning, validated through user studies across datasets.

- Proposed an intuitive approach for unsupervised end-to-end learning of image disentanglement.
- Showed that contrastive learning on image features enhances the accuracy and fidelity of generated images.

Img2Motion: Learning to Drive 3D Avatars Using Videos **05/2019 – 11/2019**
A neural motion retargeting system that animates any 3D avatar using motions extracted from 2D videos.

- Proposed a neural motion retargeting system that utilizes videos to drive 3D subjects with varying skeleton structures.
- Built a large-scale dataset with a variety of human poses by animating existing rigged human models.

K-D Tree for Faster Path Tracing **10/2019 – 11/2019**
A personal renderer that halves the running time of a traditional path tracer.

- Reproduced complex lighting environments through Monte Carlo integration.
- Used bounding boxes to represent objects and a k-d tree to store them, achieving significant runtime improvements.

Professional Service

Teaching Experience

- Teaching Assistant, USC CSCI-570: Analysis of Algorithms (Instructor: Prof. Victor Adamchik) **01/2021 – 05/2021**
- Teaching Assistant, USC CSCI-576: Multimedia Systems Design (Instructor: Prof. Parag Havaldar) **01/2022 – 05/2026**

Conference Reviewer

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) **2024 – 2026**
- ACM International Conference on Multimedia (MM) **2025 – 2026**
- AAAI Conference on Artificial Intelligence (AAAI) **2025 – 2026**
- International Conference on 3D Vision (3DV) **2025 – 2026**
- International Conference on Computer Vision (ICCV) **2025**
- Conference on Neural Information Processing Systems (NeurIPS) **2026**
- European Conference on Computer Vision (ECCV) **2026**
- IEEE International Conference on Robotics and Automation (ICRA) **2026**
- British Machine Vision Conference (BMVC) **2026**

Mentorship & Outreach

- Mentor, USC Graduate Student Mentorship Program **2025 – 2026**
- Mentor, USC Women in Engineering Mentorship Program **2025 – 2026**

Students Mentored

- **Xinjie Chen** — Master's Student, Computer Science, USC **2026 – Present**
- **Aniruth Sundararajan** — Master's Student, Computer Science, USC **2026 – Present**
- **Junsoo Kim** — Undergraduate Student, Computer Science, USC **2026 – Present**
- **Yunkai Zhan** — Undergraduate Student, Computer Science, USC **2026 – Present**

Honors & Awards

- USC Graduate Student Government (GSG) Professional Development Fund **2023, 2025**
- Outstanding Undergraduate Scholarship, Ocean University of China **2014 – 2016**

References

Prof. Yue Wang — *Assistant Professor, Computer Science, USC* yue.w@usc.edu
Prof. Ulrich Neumann — *Professor (Retired), Computer Science, USC* uneumann@usc.edu
Prof. Hao Li — *CEO, Pinscreen; Professor & Dean, MBZUAI* hao@hao-li.com
Dr. Parag Havaldar — *Part-Time Lecturer, USC; Senior Lead R&D Engineer, Blizzard* havaldar@usc.edu
Dr. Yuting Ye — *Research Director, Meta* yuting.ye@meta.com
Dr. Yuanlu Xu — *AI Research Scientist, Meta Superintelligence Labs* merayxu@gmail.com
Dr. Tony Tung — *Director of AI Research, Meta Superintelligence Labs* tony2ng@gmail.com
Dr. Chuhang Zou — *AI Research Scientist, Meta Reality Labs* zouchuhang@gmail.com
Dr. Jae Shin Yoon — *Research Scientist, Adobe Research* jaeyoon@adobe.com
Dr. Mingming He — *Senior Computer Scientist, Adobe* hmm.lillian@gmail.com